

Problem Set #1

ECO 8000
Automne 2026

Due date: October 19, 2026. Please submit by email to sam.gyetvay@gmail.com with subject ECO8000 + TP1 + your last name + student number. Late submissions will be accepted, but I might subtract some marks.

Reminder: Collaboration is allowed (and encouraged) but each student must write and submit their own solution. Remember that each problem set is only worth 10% of your final grade, while presentation + participation + final project are worth 80%. This is intentional: the goal is for you to learn, not just to get the right answer. If you get the right answer by copying from another student or from ChatGPT, you will not learn.

Instructions: I would like you to write your document using LaTeX. A popular in-browser LaTeX compiler among students and researchers is Overleaf (www.overleaf.com). This is good practice for when you have to write a paper. For the questions involving analysis of a data set, you are free to use whichever software you like. I use a combination of STATA and R in my research, and I am fairly advanced in my knowledge of both, so I can give you hints. I occasionally use Python, but I am not good at it at all. I strongly recommend against using MATLAB, SPSS, SAS, Julia or any other software. For questions that require you to write code, please include your script (STATA .do file, .R script, or other). Include comments on your code and indicate which section corresponds to each question so that I can follow it.

Use of AI: I am not going to forbid you from using AI. In my opinion, you will learn best if you try your best without AI. But if you use it in a way that helps you learn, that's fine with me. However, please beware that I have an extreme distaste for AI generated writing. Even if you used AI to help you find the solution, I strongly urge you to write your answers in your own words for all questions.

Word of encouragement: This problem set is very long, and I have given you a lot of time to work on it. I strongly encourage you to start working on it *now*. Working with data is likely the most valuable skill you will acquire in your degree, and the only way to learn it is by spending a lot of time doing it. Working with data is a very painful and frustrating process. You will most likely spend some nights screaming at your computer trying to debug some code and cursing my name. We've all been there. Just relax and try to enjoy :-)

You can find this document online at www.samgyetvay.com/teaching/eco8000/tp1_en.pdf

Question 1: Generalized Roy model (10 points) In class, we studied a version of the Roy model where workers chose option 1 (e.g. hunt) over option 0 (e.g. fish) when the earnings from 1 exceed

the earnings from 0: $Y_{i1} > Y_{i0}$. In this problem, we will consider a generalization of the Roy model where worker i chooses option 1 if $Y_{i1} - C_{i1} > Y_{i0} - C_{i0}$, where C_{i1} and C_{i0} are costs.

Question 1 (a): (1 point) We will start by defining

$$D_i = 1\{Y_{i1} - C_{i1} > Y_{i0} - C_{i0}\}$$

In words: D_i is a dummy variable equal to 1 when worker i chooses option 1. We will write earnings and costs as functions of a vector of observable worker characteristics X_i :

$$Y_{id} = \beta_d X_i + \epsilon_{id}, \quad C_{id} = \gamma_d X_i + \eta_{id}, \quad d \in \{0, 1\}$$

where $E[\epsilon_{id}|X_i] = E[\eta_{id}|X_i] = 0$. Define $\psi = (\beta_0 - \beta_1) + (\gamma_1 - \gamma_0)$ and $v_i = (\epsilon_{i1} - \epsilon_{i0}) - (\eta_{i1} - \eta_{i0})$. Show that

$$D_i = 1\{v_i > \psi X_i\}.$$

Describe this condition in words. Do workers always choose option 1 when it pays more? If not, why not?

Question 1 (b): (1 point) Show that

$$E[Y_{i1}|X_i, D_i = 1] = \beta_1 X_i + E[\epsilon_{i1}|X_i, v_i > \psi X_i]$$

What is $E[Y_{i1}|X_i, D_i = 1]$, in words?

Question 1 (c): (2 points) Suppose that

$$(\epsilon_{id}, v_i)'|X_i \sim N\left(0, \begin{bmatrix} \sigma_d^2 & \rho_d \sigma_d \\ \rho_d \sigma_d & 1 \end{bmatrix}\right), \quad d \in \{0, 1\}$$

Here σ_d is the conditional standard deviation of ϵ_{id} ; the variance of v_i is normalized to one. Let $\lambda(z) = \phi(z)/[1 - \Phi(z)]$.

Show that

$$E[Y_{i1}|X_i, D_i = 1] = \beta_1 X_i + \rho_1 \sigma_1 \lambda(\psi X_i)$$

Also show that

$$E[Y_{i0}|X_i, D_i = 0] = \beta_0 X_i - \rho_0 \sigma_0 \frac{\phi(\psi X_i)}{\Phi(\psi X_i)}$$

(Hint: use the trick from the slide titled “Conditional expectation of joint normals” to write $\epsilon_{id} = \rho_d \sigma_d v_i + e_{id}$ where $E[e_{id}|v_i, X_i] = 0$)

Question 1 (d): (2 points) Suppose we have a variable Z_i that affects costs of option 1 but not earnings so that

$$Y_{i1} = \beta_1 X_i + \epsilon_{i1}, \quad C_{i1} = \gamma_1 X_i + \varphi_1 Z_i + \eta_{i1}$$

Assume the joint normal model in part (c) also holds conditional on X_i and Z_i . We observe D_i , X_i , and Z_i for everyone, but observe Y_{i1} only when $D_i = 1$. In a Heckman two-step procedure, first estimate the probit for $D_i = 1$:

$$P(D_i = 1 \mid X_i, Z_i) = \Phi(\theta_1 X_i + \theta_2 Z_i).$$

Then, among workers with $D_i = 1$, regress observed earnings $Y_i = Y_{i1}$ on X_i and

$$\frac{\phi(\hat{\theta}_1 X_i + \hat{\theta}_2 Z_i)}{\Phi(\hat{\theta}_1 X_i + \hat{\theta}_2 Z_i)}.$$

What are θ_1 and θ_2 in terms of the model's parameters?

Question 1 (e): (2 points) Which parameters, or combinations of parameters, can be identified from the probit and the earnings regression among workers with $D_i = 1$ in part (d)? Which cannot be separately identified?

Question 1 (f): (2 points) Suppose D_i is worker i 's decision to immigrate from the home country to Canada, Y_{i1} are earnings in Canada, Y_{i0} are earnings in the home country, and C_{i1} are costs of migration. State the exclusion restriction on Z_i in the equations and in words; X_i may affect both earnings and migration costs. Give examples of X_i and a variable Z_i that affects migration costs, and explain one way your proposed Z_i might violate the restriction.

Question 2: Mulligan and Rubinstein (2008) (20 points). To answer this question, download the data set at the following link

www.samgyetvay.com/teaching/eco8000/cps.html

This is the data set analyzed by Casey Mulligan and Yonah Rubinstein in their 2008 paper “Selection, Investment, and Women’s Relative Wages Over Time.”

Question 2 (a): (1 point) We will start out by imposing the sample restrictions from the paper (described in Section III.A). Restrict the sample so it includes white, non-hispanic workers aged 26-55. Drop workers living in group quarters, and drop workers who have a missing value for the variable `married`.

Question 2 (b): (1 point) In this section we will construct the variables used in the Heckman two-step. First, create a dummy variable that identifies “full time full year” workers: workers who worked at least 50 weeks and 35 hours. (*Hint*: use the variables `wksly` and `hrslyr`). Create a measure of the real hourly wage by dividing the variable `wage_cpi` by the variable `wrkhrlr`, then create a variable equal to the log of the real hourly wage. These two variables will be the dependent variables in the probit and wage regression, respectively.

Question 2 (c): (3 points) Next we will generate the X and Z variables in the Heckman two-step (described in III.B). X contains dummies for different levels of education, marital status, region, and a fourth order polynomial (“quartic”) in potential work experience interacted with education. Start by creating a variable equal to 1 if a worker has never been married and 0 otherwise. Generate a dummy variable for each of the following education groups: high school drop outs (8 years of schooling or less), high school dropouts with 9-11 years of schooling, high school graduates, workers with some college, college graduates, advanced degrees. Generate the components of a fourth order polynomial in experience (measured in the variable `exp`) (*Hint:* to avoid floating point issues with very large, you can normalize `exp` by dividing it by 10 before squaring. I.e., generate a variable called `exp2 = (exp/10)^2`, `exp3 = (exp/10)^3` and so on.) Then create a set of variables named e.g. `exp1_hsd08`, `exp2_hsd08` for each of the education-experience interaction terms. Finally, use the variable `state` to define a variable called `region` that divides the states into northeast, midwest, south, and west¹. Z contains all the variables in X , plus an interaction between marital status and the number of children aged 0-6. The number of children aged 0-6 is contained in the variable `__age06`. Generate the interaction terms by generating variables named e.g. `nmkids` equal to the dummy variable for never married multiplied by the number of children aged 0-6. Collect all the variables in X and Z in a pair of globals named X and Z (*Hint:* if you’re not familiar with globals, you can read about them here https://comet.arts.ubc.ca/docs/5_Research/econ490-stata/04_Locals_and_Globals.html).

Question 2 (d): (2 point) The number of years of work experience is not observed in the Current Population Survey (CPS), which Mulligan and Rubinstein base their analysis on (and from which the dataset `cps.csv` is derived). The variable `exp` is a measure of potential experience. Read Appendix I of the paper and explain in your own words how the authors constructed this variable.

Question 2 (e): (3 points) We are now ready to estimate the first step. Repeat the following process, once for the period 1975-79, and once for the period 1995-99. Start by estimating by probit the relationship between full-time full year, X and Z , for women only. Use the variable `wgt` as a weight (you can use the option `[pw = wgt]`) (*Hints:* use the STATA command `probit`. Enter `help probit` into the STATA command line to read about this program. You can use the globals you created in part (c) of this question to write each probit compactly). Once you’ve estimated the probit, form the inverse Mills ratio term $\phi(\hat{b}_1 X_i + \hat{b}_2 Z_i) / \Phi(\hat{b}_1 X_i + \hat{b}_2 Z_i)$, where $\phi(z)$ is the density of a standard normal and $\Phi(z)$ is the cumulative density of a standard normal. (*Hint:* in STATA, you can use the command `predict PhiXb, p` after `probit` to generate $P(X_i \leq \hat{b}_1 X_i + \hat{b}_2 Z_i)$) and `predict Xb, xb` to generate fitted values $\hat{b}_1 X_i + \hat{b}_2 Z_i$. The command `generate fXb = normd(Xb)` allows you to calculate $\phi(\hat{b}_1 X_i + \hat{b}_2 Z_i)$). Set the value of the inverse mills ratio to zero for male workers.

Question 2 (f): (2 points) Before estimating the second-stage of the Heckit, let’s see what we

¹See https://en.wikipedia.org/wiki/List_of_regions_of_the_United_States for the list of states in each region

would get if we don't adjust for selection. For each period, run a regression of log wage on a dummy variable equal to 1 if the worker is a woman and 0 otherwise and the covariates X , restricting your sample to workers employed full time full year 50 weeks per year. Make sure to use heteroskedasticity robust standard errors. Report the coefficient on the female dummy. How do you interpret this number? How does it change between the two periods? What does this imply about the gender wage gap? Why should we be skeptical of this estimate? (*Hint*: use the command `regress` to run linear regressions. Use the option `, robust` to use heteroskedasticity robust standard errors.)

Question 2 (g): (4 points) We're now ready to estimate the second-step of the Heckit. Estimate the same regression in part (f), but this time include the inverse mills ratio variable as a control. Report the coefficients on the female dummy as well as the coefficient on the inverse mills ratio. Interpret each coefficient. How do the coefficients on the female dummy compare to the coefficients in part (f)? Are women negatively or positively selected? Why? How did this change between 1975-79 and 1995-99?

Question 2 (h): (4 points) Repeat parts (f) and (g) separately for women who are currently married and women who are not. Collect your observations in a table and comment on the results.

Question 3: Angrist and Krueger (1991) (20 points). To answer this question, download the file `NEW7080.rar` under **Angrist and Krueger (1991)** at the following link

<https://economics.mit.edu/people/faculty/josh-angrist/angrist-data-archive>

(This is a `.rar` file, which means you have to extract its contents). This is the data set analyzed by Josh Angrist and Allan Krueger in their 1991 paper "Does compulsory school attendance affect schooling and earnings?"

Question 3 (a). (2 points) This dataset is not very clean. The variable names are just `v1`, `v2`, etc. The STATA `.do` files `QOB_Table_IV.do`, `QOB_Table_V.do`, and `QOB_Table_VI.do` at the link above contain some STATA code that will identify variable names. Use it to clean up the data. Try to guess what each variable represents based on its name. It might be helpful to briefly read the paper.

Question 3 (b). (1 point) The variable `YOB` contains the year of birth of each individual in the data set. However, this variable is not coded consistently across years. Transform the variable so that it is consistent. (*Hint*: Use the STATA command `tabulate YOB` to summarize the values).

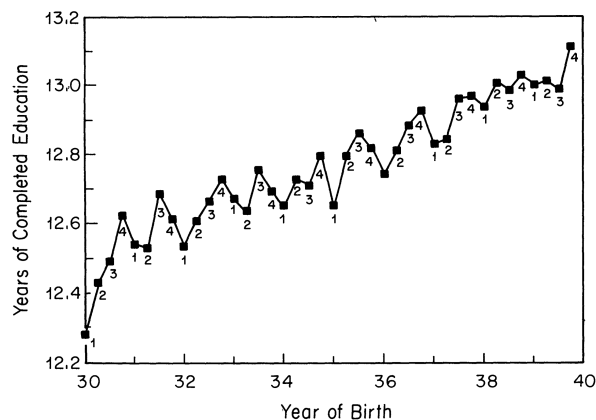


FIGURE I
Years of Education and Season of Birth
1980 Census
Note. Quarter of birth is listed below each observation.

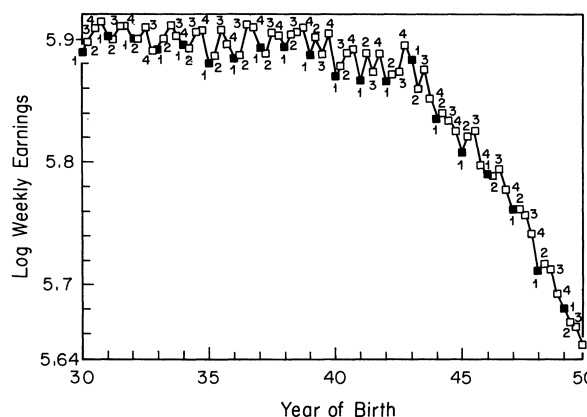


FIGURE V
Mean Log Weekly Wage, by Quarter of Birth
All Men Born 1930–1949; 1980 Census

Question 3 (c). (5 points) Re-create Figures I, II, and V from Angrist and Krueger (1991). Try to make the figure as similar as the one in the paper as possible. Explain how the pattern in the figures relate to the identification strategy in the paper. (*Hint:* you can combine the two variables YOB and QOB by creating a new variable equal to $YOB + QOB/4$). (*Hint:* use the STATA commands `twoway line` and `twoway scatter`).

Question 3 (d). (3 points) The data set contains the 1970 census as well as the 1980 census. Use these observations to create a version of Figure I-II and Figure V that for workers born between 1920 and 1929. (*Hint:* the variable `LWKLYWGE` has missing values for all observations for workers born in 1920-29. However, one of the unnamed variables (i.e., one of `v14`, `v15`, `v17`, etc.) contains non-missing values.)

Question 3 (e). (5 points) Reproduce Table I from the paper. Don't worry about making it look exactly like the one in the paper. (*Hint:* use the command `summarize` to calculate means. For the differences in column (3), use the command `regress` and regress the outcome on a dummy variable equal to 1 if the worker was born in the first quarter). You don't have to calculate the standard errors for the Wald estimate, but I will give bonus marks if you can figure it out and if you obtain identical standard errors to those in the paper.

Question 3 (f). (4 points) Repeat the analysis in Panel B of Table II, but calculate estimates separately for individuals born in 1930-39 and individuals born in 1940-49. Comment on your findings.

Question 4: LATE theorem. (20 points) In this problem, you will explore the LATE theorem, first proved by Joshua Angrist and Guido Imbens in their 1994 paper "Identification and estimation of local average treatment effects" through a hypothetical example. An education non-profit has teamed up with labor economists at UQAM to test a program aimed at increasing awareness about

tuition subsidies for students from low-income families. Together, the research team have put together a sample of 1000 young adults aged 18-21 whose parental income is below 150% of the poverty line. The researchers randomly assigned each student i into the treatment group ($Z_i = 1$) or the control group ($Z_i = 0$). Students in the treatment group receive an information package telling them about tuition subsidies they are eligible for and detailed instructions on how to apply. Let D_i be a dummy variable equal to 1 if i enrolls in a program and 0 otherwise. And let Y_i be i 's income 5 years later.

Question 4 (a). (5 point) Let $D_i(1)$ denote worker i 's enrollment if they receive information, and $D_i(0)$ denote their enrollment decision if they didn't receive information. Describe in words the following groups, in the context of this application

- $1 = D_i(1) > D_i(0) = 0$
- $D_i(1) = D_i(0) = 1$
- $D_i(1) = D_i(0) = 0$

Question 4 (b). (5 point) State formally the three assumptions of the LATE theorem (from the slides) and explain them in your own words in the context of this application. For each assumption, explain why it is satisfied (or not) for this application.

Question 4 (c). (5 point) Prove the LATE theorem. That is, prove that, under the assumptions you stated in part (b),

$$\frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]} = E[Y_i(1) - Y_i(0)|D_i(1) > D_i(0)]$$

Question 4 (d). (5 point) Do you expect the LATE to be larger, smaller, or the same as the ATE in this application? Explain your answer.